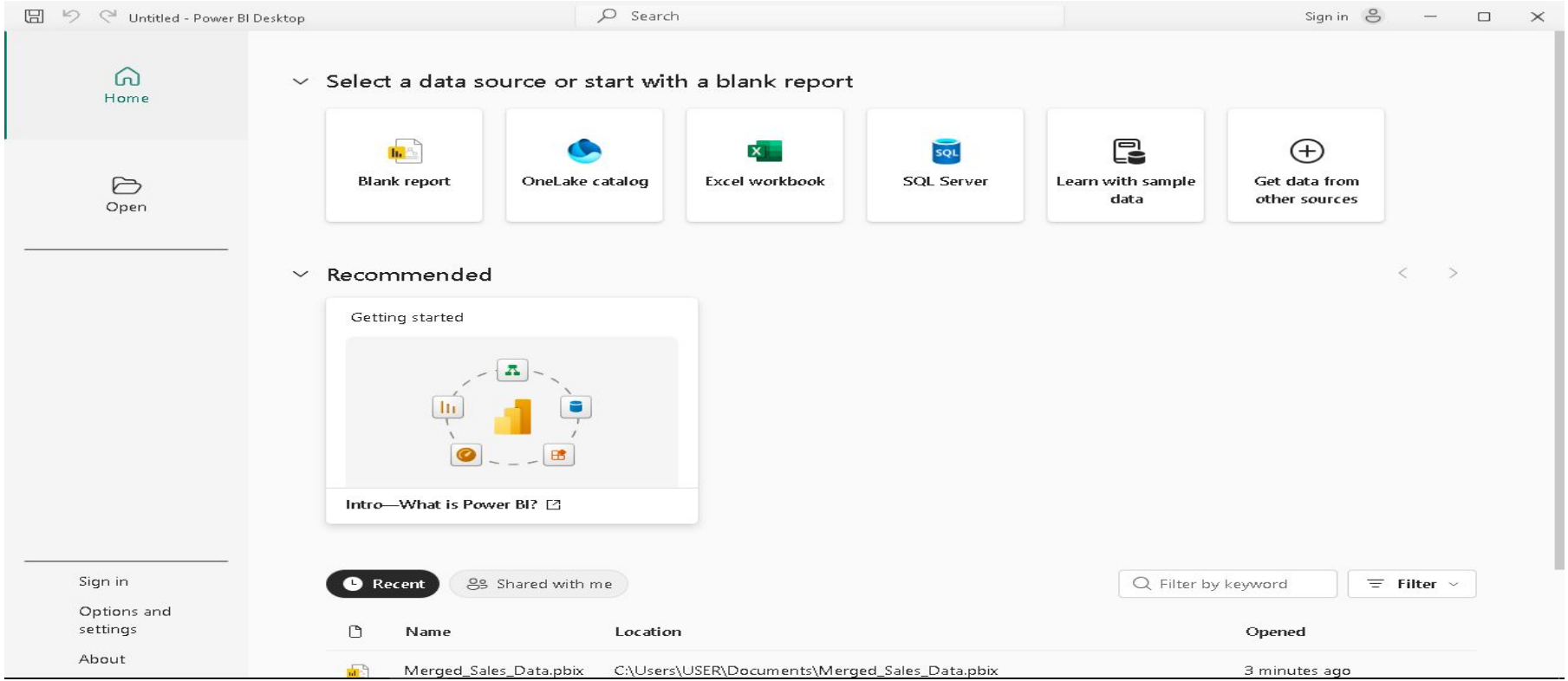
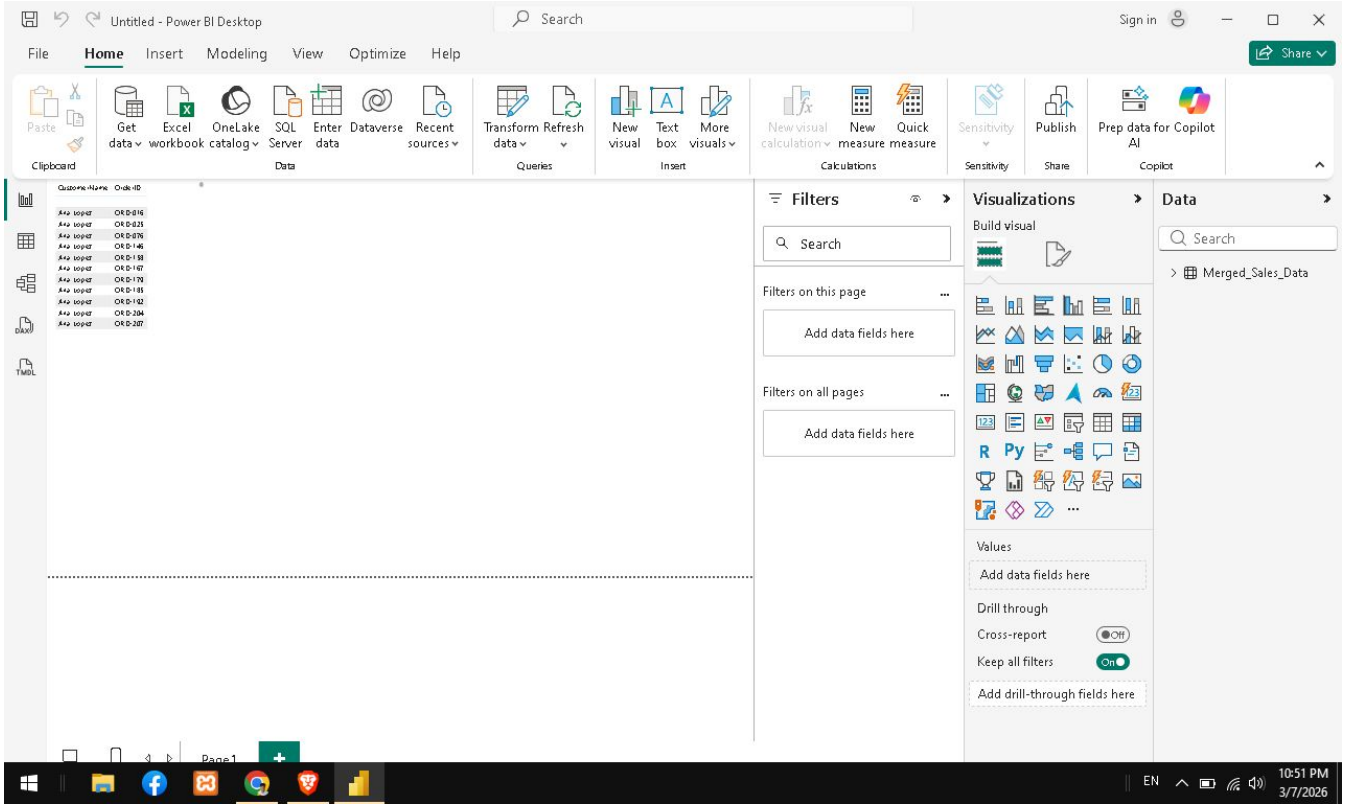
Merged Data, DAX, and DAX Function Insights

Kenneth A. Camasura
Instructor: Joseph A. Vistal
IS 107 - Business Intelligence

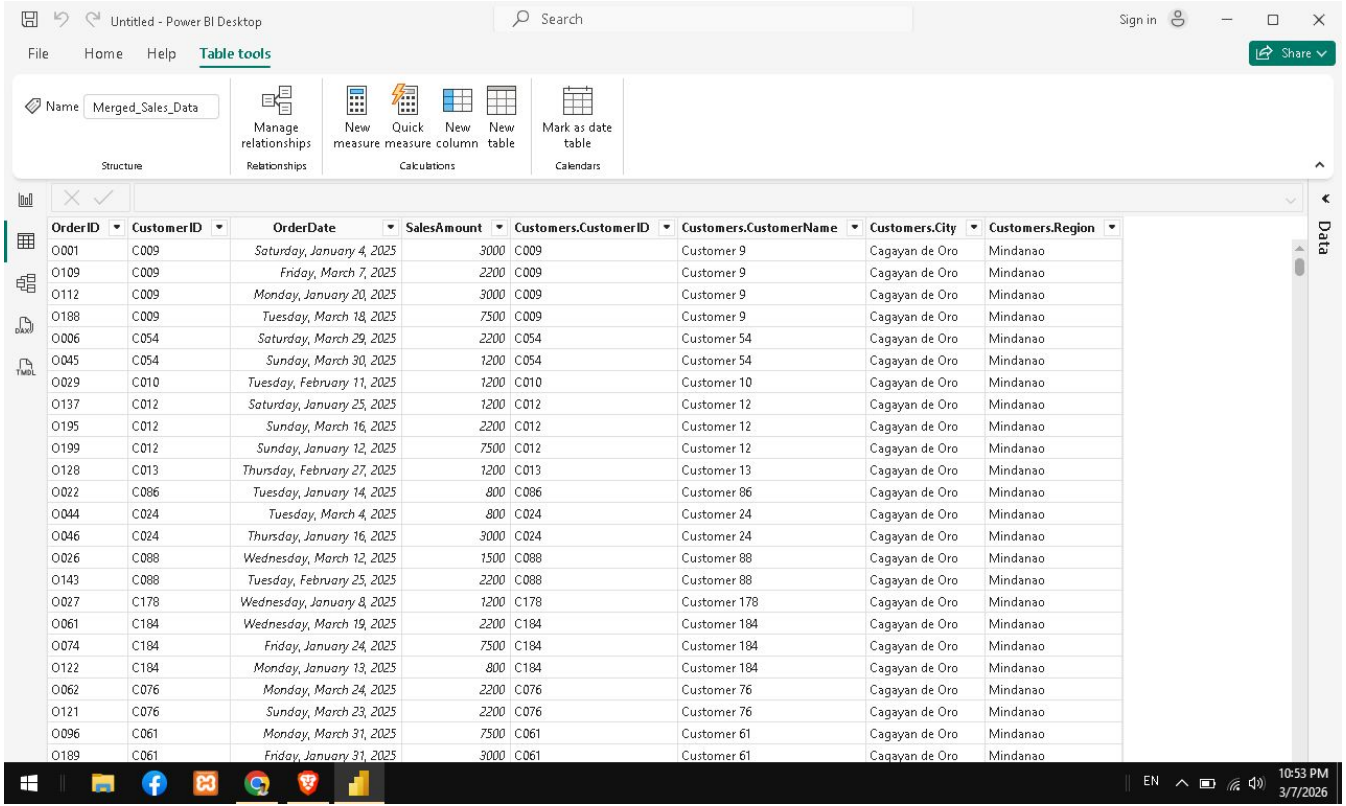
Part 1: Data Modeling Fundamentals Using Merged Data | Step 1 Open Power BI Desktop.



Part 1: Data Modeling Fundamentals Using Merged Data | Step 2 Load the Merged_Sales_Data table.



Part 1: Data Modeling Fundamentals Using Merged Data| Step 3 Switch to Data View.



OrderID	CustomerID	OrderDate	SalesAmount	Customers.CustomerID	Customers.CustomerName	Customers.City	Customers.Region
O001	C009	Saturday, January 4, 2025	3000	C009	Customer 9	Cagayan de Oro	Mindanao
O109	C009	Friday, March 7, 2025	2200	C009	Customer 9	Cagayan de Oro	Mindanao
O112	C009	Monday, January 20, 2025	3000	C009	Customer 9	Cagayan de Oro	Mindanao
O188	C009	Tuesday, March 18, 2025	7500	C009	Customer 9	Cagayan de Oro	Mindanao
O006	C054	Saturday, March 29, 2025	2200	C054	Customer 54	Cagayan de Oro	Mindanao
O045	C054	Sunday, March 30, 2025	1200	C054	Customer 54	Cagayan de Oro	Mindanao
O029	C010	Tuesday, February 11, 2025	1200	C010	Customer 10	Cagayan de Oro	Mindanao
O137	C012	Saturday, January 25, 2025	1200	C012	Customer 12	Cagayan de Oro	Mindanao
O195	C012	Sunday, March 16, 2025	2200	C012	Customer 12	Cagayan de Oro	Mindanao
O199	C012	Sunday, January 12, 2025	7500	C012	Customer 12	Cagayan de Oro	Mindanao
O128	C013	Thursday, February 27, 2025	1200	C013	Customer 13	Cagayan de Oro	Mindanao
O022	C086	Tuesday, January 14, 2025	800	C086	Customer 86	Cagayan de Oro	Mindanao
O044	C024	Tuesday, March 4, 2025	800	C024	Customer 24	Cagayan de Oro	Mindanao
O046	C024	Thursday, January 16, 2025	3000	C024	Customer 24	Cagayan de Oro	Mindanao
O026	C088	Wednesday, March 12, 2025	1500	C088	Customer 88	Cagayan de Oro	Mindanao
O143	C088	Tuesday, February 25, 2025	2200	C088	Customer 88	Cagayan de Oro	Mindanao
O027	C178	Wednesday, January 8, 2025	1200	C178	Customer 178	Cagayan de Oro	Mindanao
O061	C184	Wednesday, March 19, 2025	2200	C184	Customer 184	Cagayan de Oro	Mindanao
O074	C184	Friday, January 24, 2025	7500	C184	Customer 184	Cagayan de Oro	Mindanao
O122	C184	Monday, January 13, 2025	800	C184	Customer 184	Cagayan de Oro	Mindanao
O062	C076	Monday, March 24, 2025	2200	C076	Customer 76	Cagayan de Oro	Mindanao
O121	C076	Sunday, March 23, 2025	2200	C076	Customer 76	Cagayan de Oro	Mindanao
O096	C061	Monday, March 31, 2025	7500	C061	Customer 61	Cagayan de Oro	Mindanao
O189	C061	Friday, January 31, 2025	3000	C061	Customer 61	Cagayan de Oro	Mindanao

Guide Questions

1. Is the dataset in flat-table format?

Answer: Yes

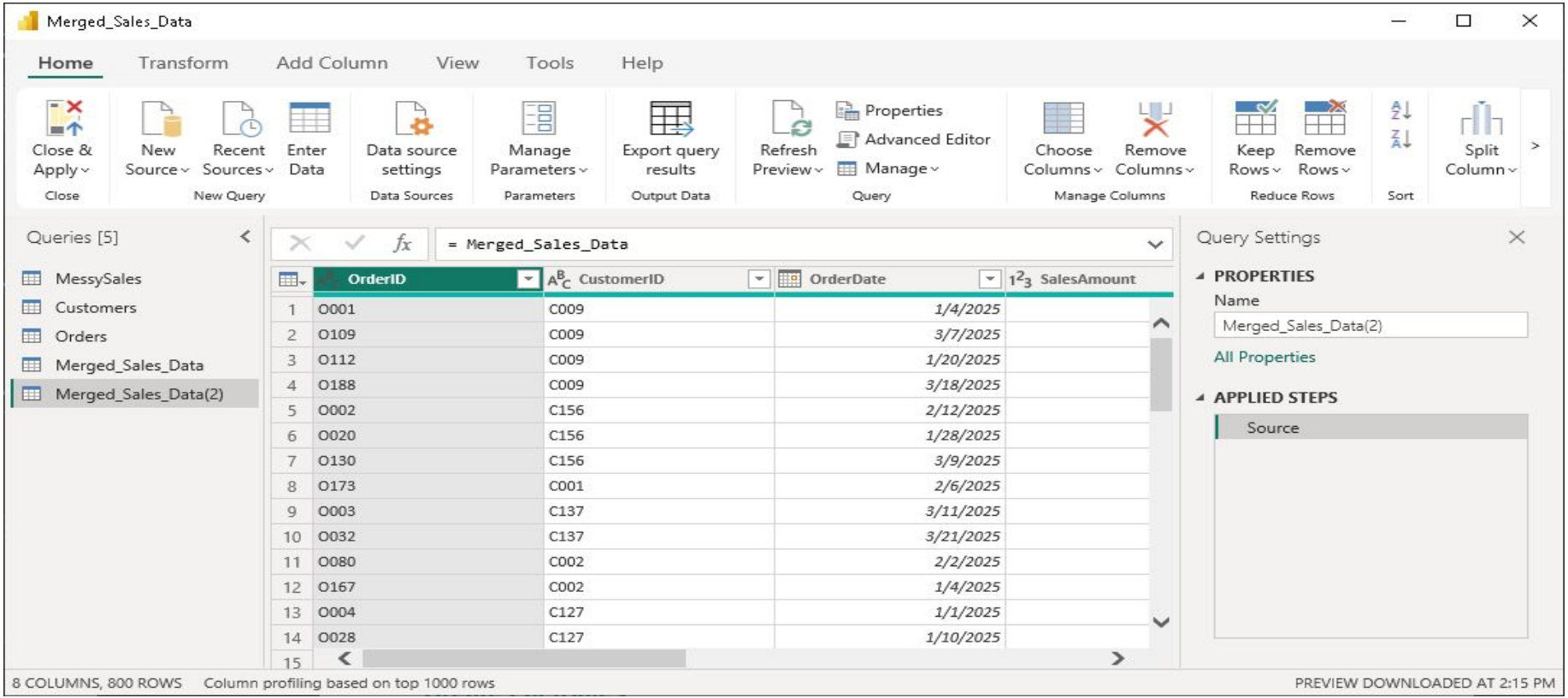
2. Which columns describe customers?

Answer: Customers.CustomerName,

3. Which columns represent transactions?

Answer: SalesAmount

Part 2: Identify Fact and Dimension Tables | Create DimCustomer Step 1: Right-click table → Reference.



Part 2: Identify Fact and Dimension Tables | Create DimCustomer Step 2: Remove columns not related to customers.

Merged_Sales_Data

Home | Transform | Add Column | View | Tools | Help

Close & Apply | New Source | Recent Sources | Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Properties | Advanced Editor | Manage | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Split Column

Close | New Query | Data Sources | Parameters | Output Data | Query | Manage Columns | Reduce Rows | Sort

Queries [5]

- MessySales
- Customers
- Orders
- Merged_Sales_Data
- Merged_Sales_Data(2)

```
= Table.RemoveColumns(Source,{"OrderDate", "SalesAmount", "OrderID",
```

	Customers.CustomerID	Customers.CustomerName	Customers.City	Customers.
1	C009	Customer 9	Cagayan de Oro	Mindanao
2	C009	Customer 9	Cagayan de Oro	Mindanao
3	C009	Customer 9	Cagayan de Oro	Mindanao
4	C009	Customer 9	Cagayan de Oro	Mindanao
5	C156	Customer 156	Dumaguete	Visayas
6	C156	Customer 156	Dumaguete	Visayas
7	C156	Customer 156	Dumaguete	Visayas
8	C001	Customer 1	Legazpi	Luzon
9	C137	Customer 137	Davao	Mindanao
10	C137	Customer 137	Davao	Mindanao
11	C002	Customer 2	Manila	Luzon
12	C002	Customer 2	Manila	Luzon
13	C127	Customer 127	Tacloban	Visayas
14	C127	Customer 127	Tacloban	Visayas
15				

Query Settings

PROPERTIES

Name: Merged_Sales_Data(2)

All Properties

APPLIED STEPS

- Source
- Removed Columns

4 COLUMNS, 800 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 2:15 PM

Part 2: Identify Fact and Dimension Tables | Create DimCustomer Step 3: Remove duplicates based on CustomerID.

Merged_Sales_Data

Home | Transform | Add Column | View | Tools | Help

Close & Apply | New Source | Recent Sources | Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Properties | Advanced Editor | Manage | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Split Column

Close | New Query | Data Sources | Parameters | Output Data | Query | Manage Columns | Reduce Rows | Sort

Queries [5]

- MessySales
- Customers
- Orders
- Merged_Sales_Data
- Merged_Sales_Data(2)

```
= Table.Distinct(#"Removed Columns", {"Customers.CustomerID"})
```

	Customers.CustomerID	Customers.CustomerName	Customers.City	Customers.
1	C009	Customer 9	Cagayan de Oro	Mindanao
2	C156	Customer 156	Dumaguete	Visayas
3	C001	Customer 1	Legazpi	Luzon
4	C137	Customer 137	Davao	Mindanao
5	C002	Customer 2	Manila	Luzon
6	C127	Customer 127	Tacloban	Visayas
7	C003	Customer 3	Butuan	Mindanao
8	C105	Customer 105	Butuan	Mindanao
9	C004	Customer 4	Dumaguete	Visayas
10	C054	Customer 54	Cagayan de Oro	Mindanao
11	C091	Customer 91	Manila	Luzon
12	C016	Customer 16	Butuan	Mindanao
13	C007	Customer 7	Zamboanga	Mindanao
14	C027	Customer 27	Baguio	Luzon
15				

Query Settings

PROPERTIES

Name: Merged_Sales_Data(2)

All Properties

APPLIED STEPS

- Source
- Removed Columns
- Removed Duplicates

4 COLUMNS, 178 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 2:17 PM

Part 2: Identify Fact and Dimension Tables | Create DimCustomer Step 4: Rename table to DimCustomer.

Merged_Sales_Data

Home | Transform | Add Column | View | Tools | Help

Close & Apply | New Source | Recent Sources | Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Properties | Advanced Editor | Manage | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Split Column

Close | New Query | Data Sources | Parameters | Output Data | Query | Manage Columns | Reduce Rows | Sort

Queries [5]

- MessySales
- Customers
- Orders
- Merged_Sales_Data
- DimCustomer

```
= Table.Distinct(#"Removed Columns", {"Customers.CustomerID"})
```

	Customers.CustomerID	Customers.CustomerName	Customers.City	Customers.
3	C001	Customer 1	Legazpi	Luzon
4	C137	Customer 137	Davao	Mindanao
5	C002	Customer 2	Manila	Luzon
6	C127	Customer 127	Tacloban	Visayas
7	C003	Customer 3	Butuan	Mindanao
8	C105	Customer 105	Butuan	Mindanao
9	C004	Customer 4	Dumaguete	Visayas
10	C054	Customer 54	Cagayan de Oro	Mindanao
11	C091	Customer 91	Manila	Luzon
12	C016	Customer 16	Butuan	Mindanao
13	C007	Customer 7	Zamboanga	Mindanao
14	C027	Customer 27	Baguio	Luzon
15	C094	Customer 94	Dumaguete	Visayas
16	C010	Customer 10	Cagayan de Oro	Mindanao
17				

Query Settings

PROPERTIES

Name: DimCustomer

All Properties

APPLIED STEPS

- Source
- Removed Columns
- Removed Duplicates

4 COLUMNS, 178 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 2:18 PM

Part 2: Identify Fact and Dimension Tables | Create FactSales Step 1: Reference original table again.

Merged_Sales_Data

Home | Transform | Add Column | View | Tools | Help

Close & Apply | New Source | Recent Sources | Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Properties | Advanced Editor | Manage | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Split Column

Close | New Query | Data Sources | Parameters | Output Data | Query | Manage Columns | Reduce Rows | Sort

Queries [6]

- MessySales
- Customers
- Orders
- Merged_Sales_Data
- DimCustomer
- Merged_Sales_Data (2)

= Merged_Sales_Data

	OrderID	CustomerID	OrderDate	SalesAmount
1	O001	C009	1/4/2025	
2	O109	C009	3/7/2025	
3	O112	C009	1/20/2025	
4	O188	C009	3/18/2025	
5	O002	C156	2/12/2025	
6	O020	C156	1/28/2025	
7	O130	C156	3/9/2025	
8	O173	C001	2/6/2025	
9	O003	C137	3/11/2025	
10	O032	C137	3/21/2025	
11	O080	C002	2/2/2025	
12	O167	C002	1/4/2025	
13	O004	C127	1/1/2025	
14	O028	C127	1/10/2025	
15				

Query Settings

PROPERTIES

Name

Merged_Sales_Data (2)

All Properties

APPLIED STEPS

Source

8 COLUMNS, 800 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 2:19 PM

Part 2: Identify Fact and Dimension Tables | Create FactSales Step 2: Keep only transaction-related columns.

Merged_Sales_Data

Home | Transform | Add Column | View | Tools | Help

Close & Apply | New Source | Recent Sources | Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Properties | Advanced Editor | Manage | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Split Column

Close | New Query | Data Sources | Parameters | Output Data | Query | Manage Columns | Reduce Rows | Sort

Queries [6]

- MessySales
- Customers
- Orders
- Merged_Sales_Data
- DimCustomer
- Merged_Sales_Data (2)

```
= Table.RemoveColumns(Source,{"Customers.CustomerID",
```

	OrderID	CustomerID	OrderDate	SalesAmount
1	O001	C009	1/4/2025	
2	O109	C009	3/7/2025	
3	O112	C009	1/20/2025	
4	O188	C009	3/18/2025	
5	O002	C156	2/12/2025	
6	O020	C156	1/28/2025	
7	O130	C156	3/9/2025	
8	O173	C001	2/6/2025	
9	O003	C137	3/11/2025	
10	O032	C137	3/21/2025	
11	O080	C002	2/2/2025	
12	O167	C002	1/4/2025	
13	O004	C127	1/1/2025	
14	O028	C127	1/10/2025	
15				

Query Settings

PROPERTIES

Name: Merged_Sales_Data (2)

All Properties

APPLIED STEPS

- Source
- Removed Columns

4 COLUMNS, 800 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 2:19 PM

Part 2: Identify Fact and Dimension Tables | Create FactSales Step 3: Rename to FactSales.

Merged_Sales_Data

Home | Transform | Add Column | View | Tools | Help

Close & Apply | New Source | Recent Sources | Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Properties | Advanced Editor | Manage | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Split Column

Close | New Query | Data Sources | Parameters | Output Data | Query | Manage Columns | Reduce Rows | Sort

Queries [6]

- MessySales
- Customers
- Orders
- Merged_Sales_Data
- DimCustomer
- FactSales

```
= Table.RemoveColumns(Source,{"Customers.CustomerID",
```

	OrderID	CustomerID	OrderDate	SalesAmount
1	O001	C009	1/4/2025	
2	O109	C009	3/7/2025	
3	O112	C009	1/20/2025	
4	O188	C009	3/18/2025	
5	O002	C156	2/12/2025	
6	O020	C156	1/28/2025	
7	O130	C156	3/9/2025	
8	O173	C001	2/6/2025	
9	O003	C137	3/11/2025	
10	O032	C137	3/21/2025	
11	O080	C002	2/2/2025	
12	O167	C002	1/4/2025	
13	O004	C127	1/1/2025	
14	O028	C127	1/10/2025	
15				

Query Settings

PROPERTIES

Name: FactSales

All Properties

APPLIED STEPS

- Source
- Removed Columns

4 COLUMNS, 800 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 2:20 PM

Guide Questions

1.Why must dimension tables contain unique keys?

In Power BI, dimension tables must contain unique keys because they serve as the reference tables that connect to fact tables through relationships in the data model. A unique key ensures that each row in the dimension table represents only one specific entity, such as a single customer in the DimCustomer table. When the key is unique, Power BI can correctly create a one-to-many relationship between the dimension table (one side) and the fact table (many side). This allows filters, slicers, and visuals in reports to work properly, ensuring that the data is accurately filtered and aggregated when users analyze the report.

Guide Questions

2. What problems occur if duplicates exist in DimCustomer?

If duplicates exist in the DimCustomer table in Power BI, the relationship between the dimension table and the fact table can become incorrect or invalid. Power BI may not allow the table to be placed on the “one” side of a relationship because the key is no longer unique. This can lead to issues such as ambiguous relationships, incorrect filtering, or inaccurate totals in reports. For example, when a user filters by a specific customer, the duplicates may cause transactions to be counted multiple times or produce inconsistent results in visuals. As a result, duplicates in DimCustomer can reduce the reliability and accuracy of the Power BI data model and the insights generated from it.

Part 3: Define Primary & Foreign Keys

Identify Keys

Table	Key	Type
DimCustomer	CustomerID	Primary Key
FactSales	CustomerID	Foreign Key
FactSales	OrderID	Primary Key

Guide Questions

1. What is a primary key?

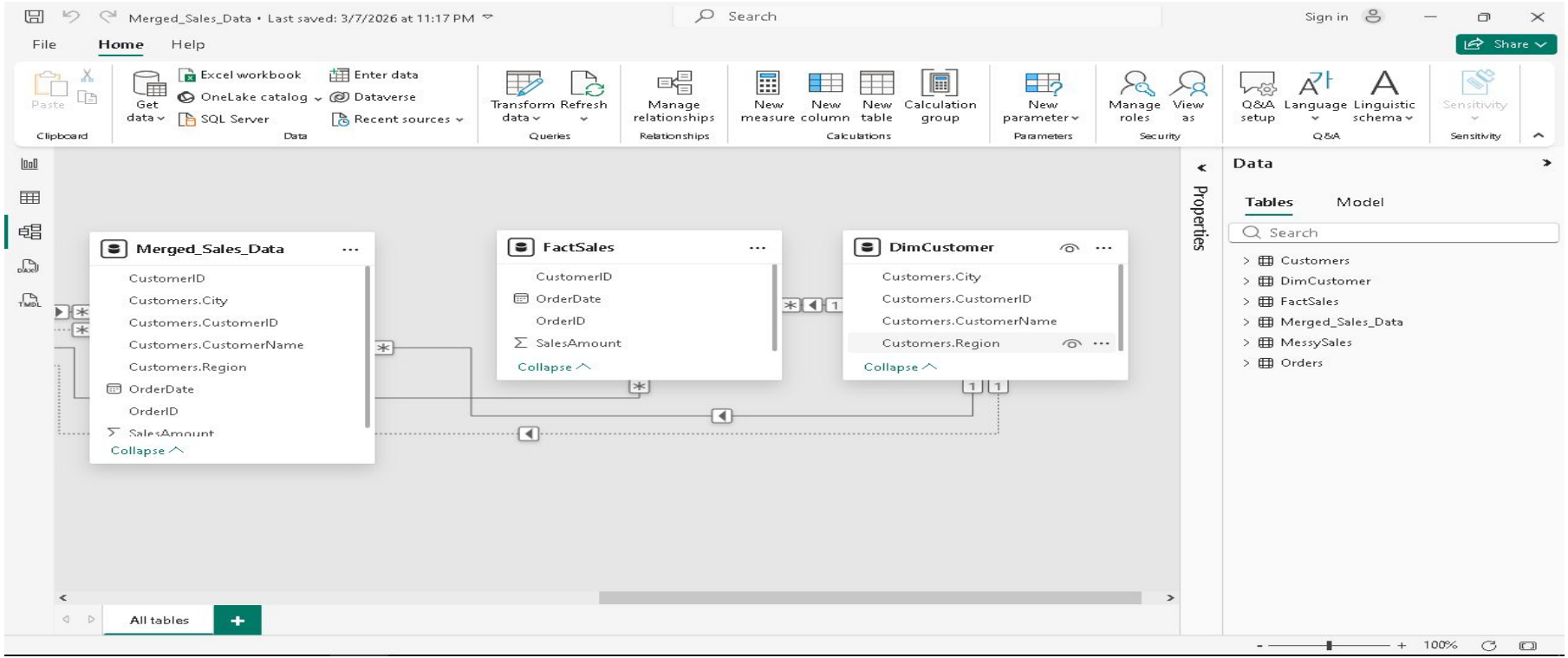
A primary key is a column or a set of columns in a table that uniquely identifies each record in that table. It ensures that every row has a distinct value and that no duplicate or null values exist in that column. In data modeling and analytics tools such as Microsoft Power BI, primary keys are commonly used in dimension tables to maintain data integrity and to establish clear relationships with other tables in the data model. By having a unique identifier, Power BI can correctly match records between tables, allowing accurate filtering, aggregation, and analysis in reports and dashboards.

Guide Questions

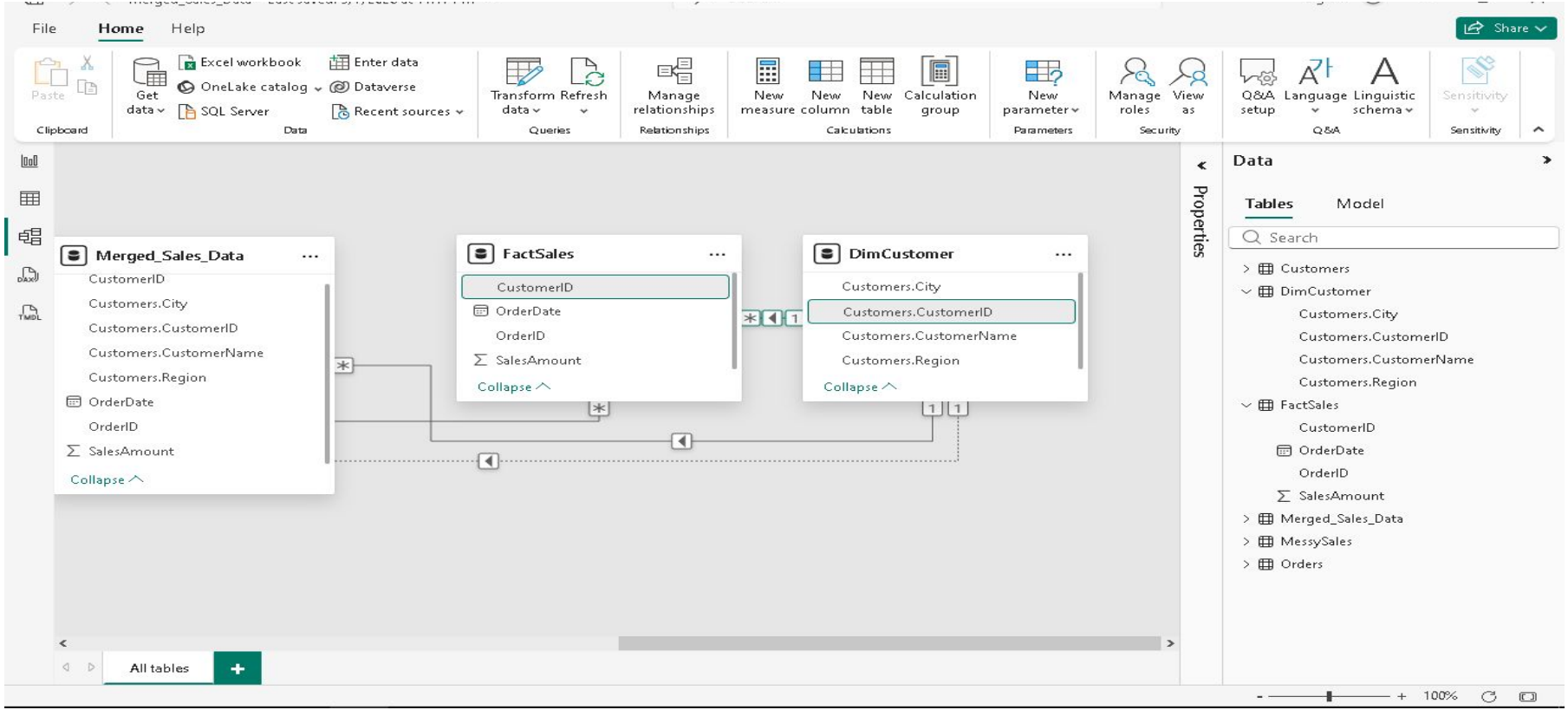
2. Why is CustomerID a foreign key in FactSales?

CustomerID is a foreign key in the FactSales table because it references the primary key in the DimCustomer table. This means that each sales transaction stored in FactSales is linked to a specific customer record in DimCustomer. In Microsoft Power BI, this relationship enables a one-to-many connection where the customer table represents the “one” side and the sales table represents the “many” side. As a result, when users filter or analyze data by customer in Power BI reports, the system can correctly associate sales transactions with the corresponding customer details, allowing accurate analysis of customer purchasing behavior and sales performance.

Part 4: Create Relationships | Step 1: Go to Model View.

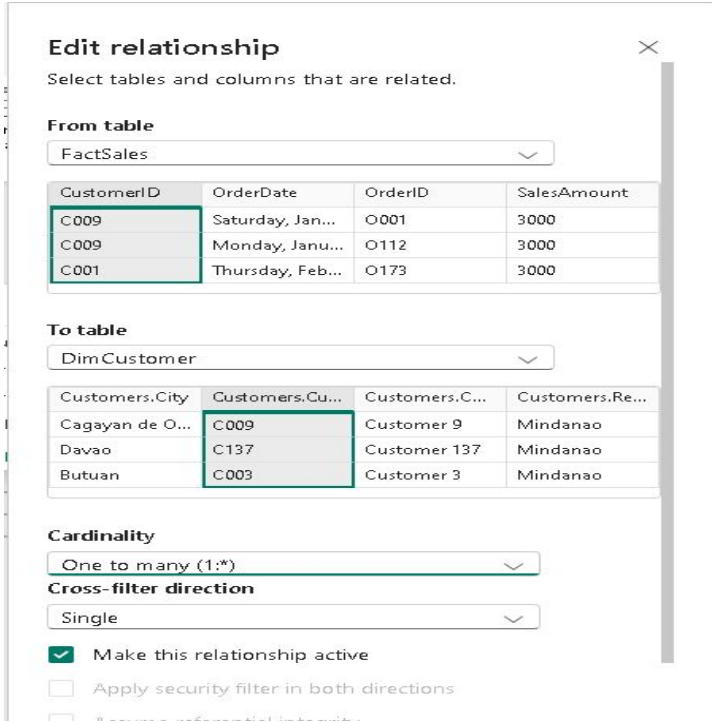


Part 4: Create Relationships | Step 2: Drag CustomerID from FactSales → DimCustomer.



Part 4: Create Relationships | Ensure relationship:

One-to-Many (Dim → Fact), Single direction filter



Guide Questions

1. Why should relationships flow from dimension to fact?

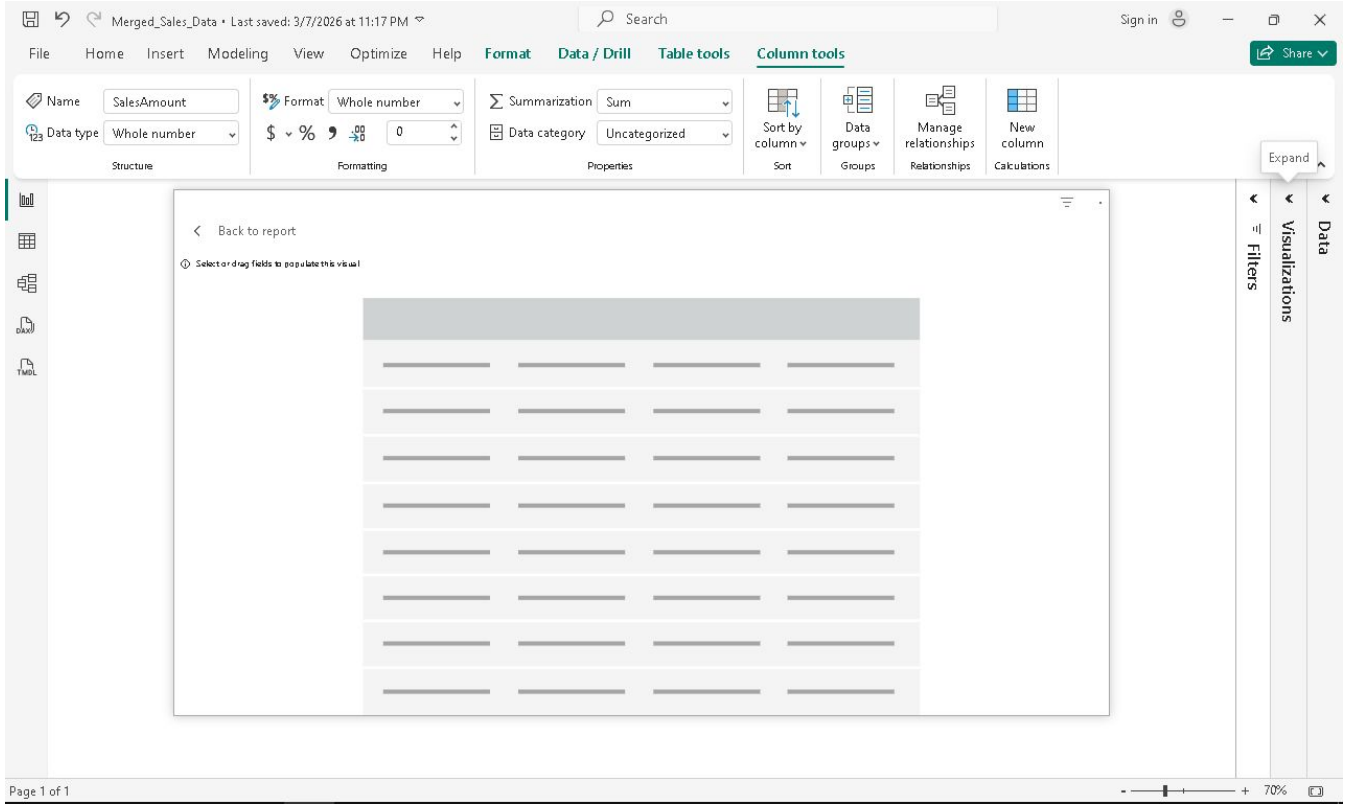
In data modeling for Microsoft Power BI, relationships should flow from the dimension table to the fact table because dimension tables contain descriptive information used to filter and categorize data, while fact tables store transactional or measurable data such as sales or quantities. When the relationship flows from dimension to fact, filters applied to attributes in the dimension table—such as customer name, product category, or date—properly propagate to the fact table. This ensures that reports, slicers, and visualizations display accurate and consistent results during analysis.

Guide Questions

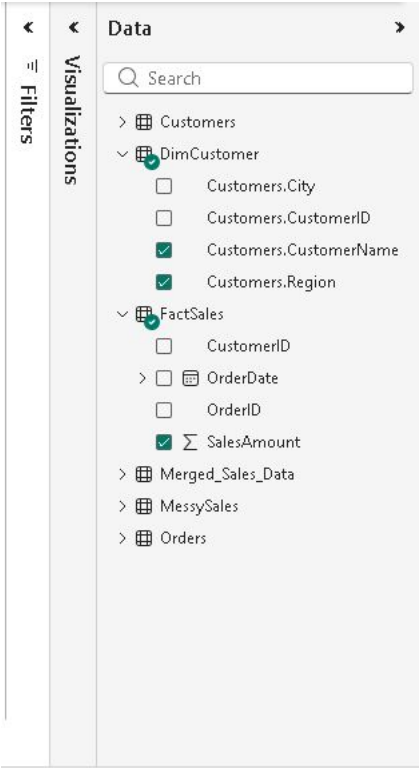
2. What happens if the relationship is Many-to-Many?

If a relationship in Microsoft Power BI is Many-to-Many, it means that multiple records in one table can relate to multiple records in another table. This can lead to ambiguity in filtering and calculations because Power BI may not clearly determine how records should be matched between the tables. As a result, measures and aggregations may produce incorrect totals or unexpected results. Many-to-many relationships can also make the data model more complex and harder to maintain, which is why the recommended design is usually a one-to-many relationship between dimension and fact tables.

Part 5: Validate the Data Model | Step 1 Create a Table visual.



Part 5: Validate the Data Model | Step 2 Add: CustomerName, Region, SalesAmount



Part 5: Validate the Data Model | Step 3 Verify that data aggregates correctly.

Back to report

Customers.CustomerName	Customers.Region	Sum of SalesAmount
Customer 98	Luzon	91600
Customer 96	Visayas	14800
Customer 95	Visayas	15000
Customer 94	Visayas	10800
Customer 93	Visayas	96000
Customer 92	Luzon	7600
Customer 91	Luzon	189800
Customer 90	Mindanao	16800
Customer 9	Mindanao	48800
Customer 89	Luzon	186800
Customer 88	Mindanao	101800
Customer 87	Mindanao	15000
Customer 86	Mindanao	1600
Customer 85	Visayas	4000
Customer 84	Luzon	91600
Customer 82	Luzon	90000
Customer 81	Visayas	186000
Customer 80	Luzon	30000
Customer 8	Luzon	94400
Customer 79	Visayas	13600
Customer 78	Mindanao	6000
Customer 77	Luzon	9000
Customer 76	Mindanao	11200
Customer 75	Mindanao	7400
Customer 74	Visayas	16600
Customer 73	Visayas	15000
Customer 72	Luzon	7600
Customer 71	Luzon	90000
Customer 7	Mindanao	30000
Total		7788800

Guide Questions

1. Did sales aggregate correctly per customer?

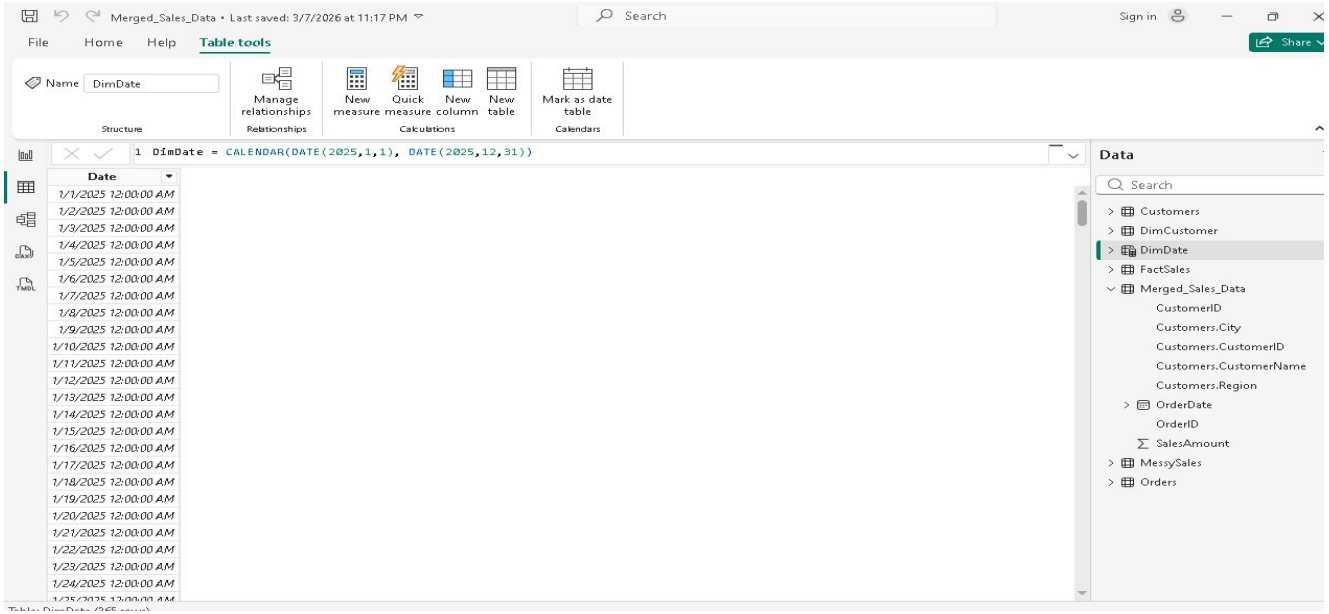
Answer: Yes

2. What would happen if relationships were missing?

Answer: Then, it will not be aggregate the data of the two table.

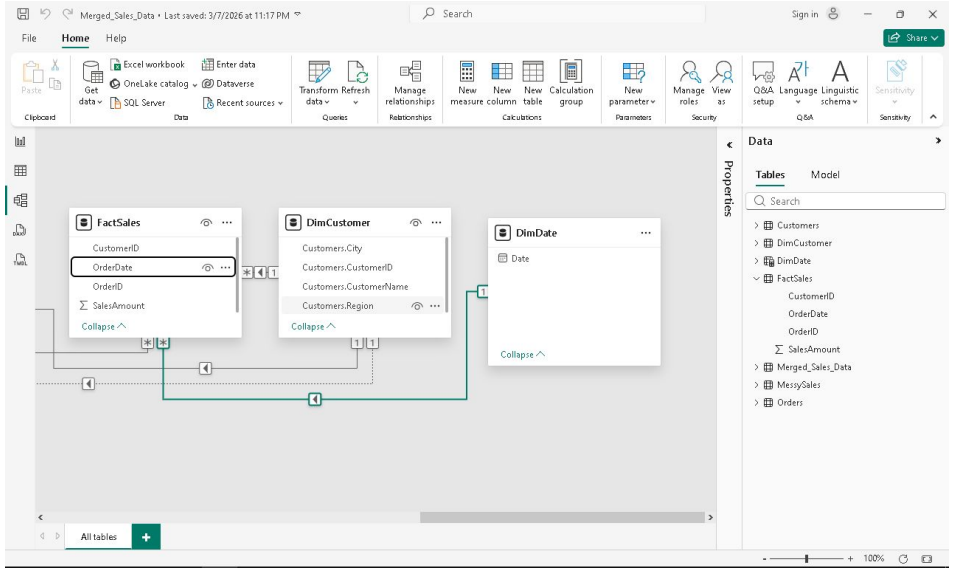
Enhancement Activities | Enhancement 1: Add DimDate Table

Create a Date table using:
DimDate = CALENDAR(DATE(2025,1,1), DATE(2025,12,31))



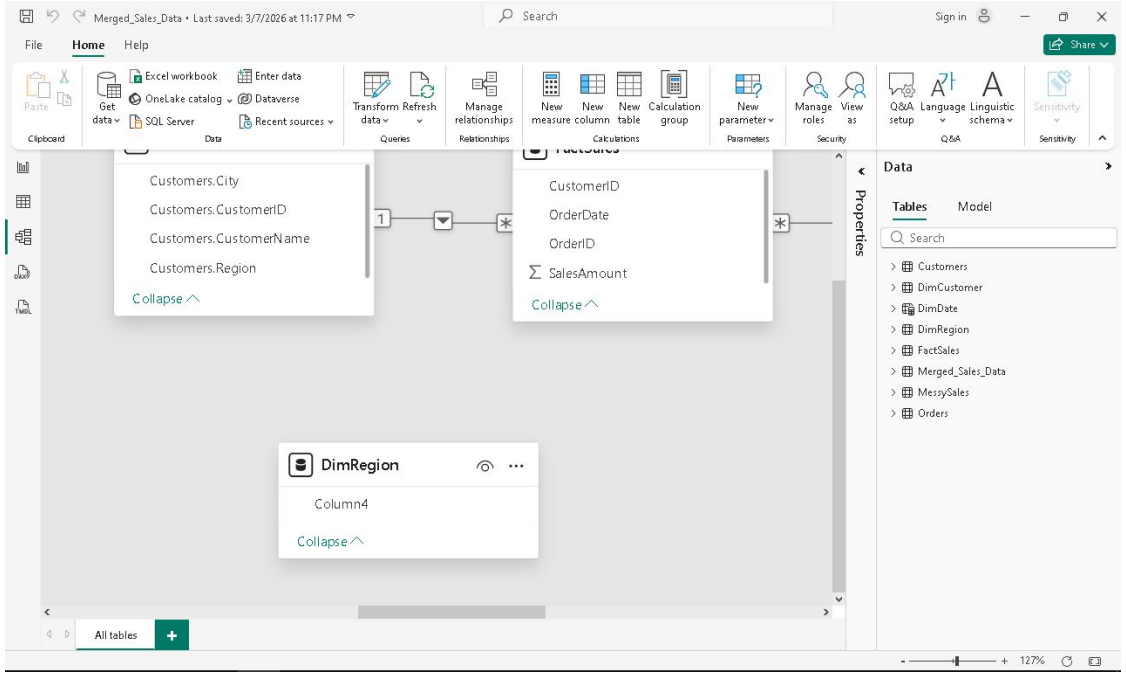
Enhancement Activities | Enhancement 1: Add DimDate Table

Relate:
FactSales[OrderDate] → DimDate[Date]



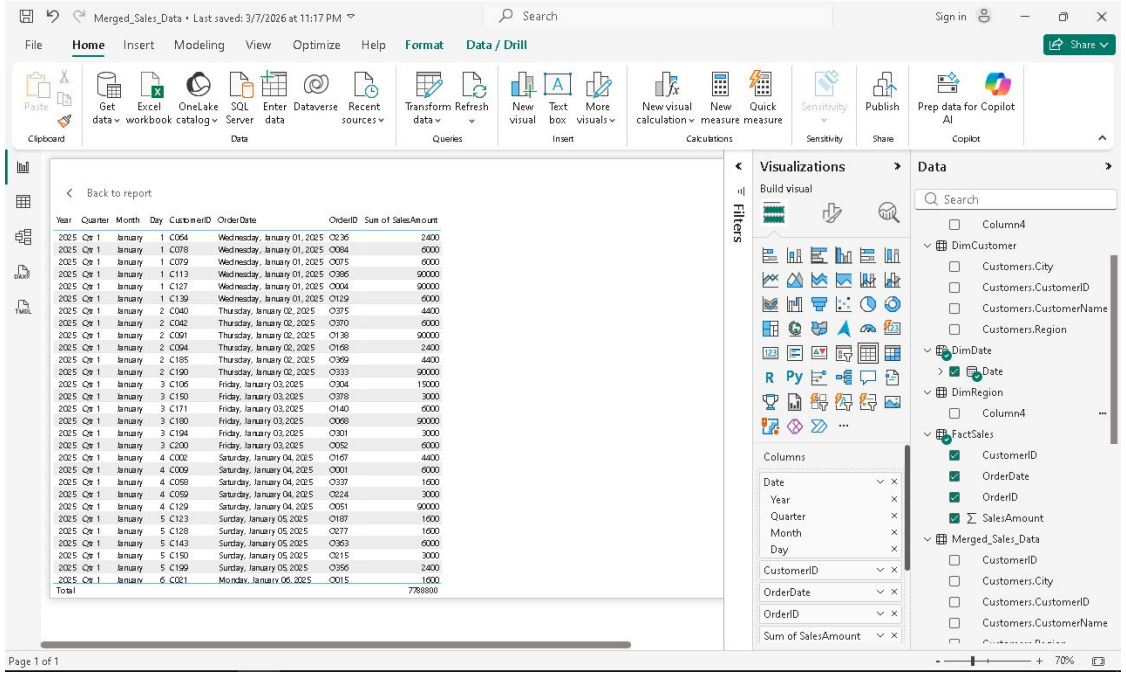
Enhancement Activities | Enhancement 2: Create DimRegion Table

Separate regions into a dimension for advanced modeling.



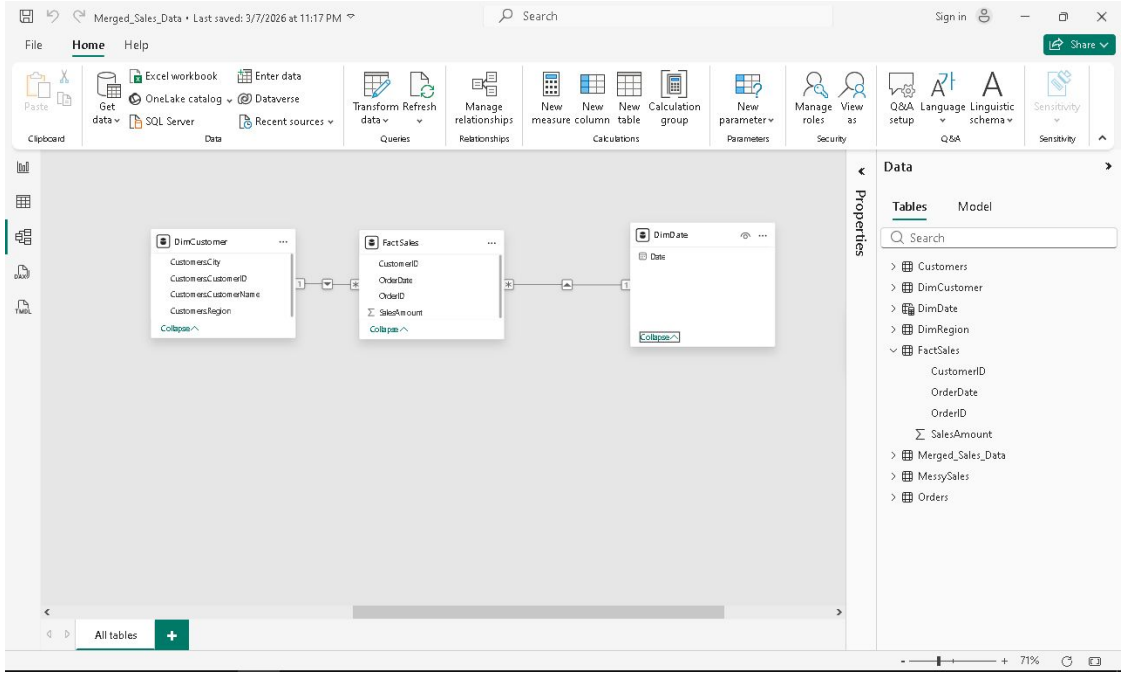
Enhancement Activities | Enhancement 3: Compare Flat vs Star Schema Performance

Create visuals using a flat table

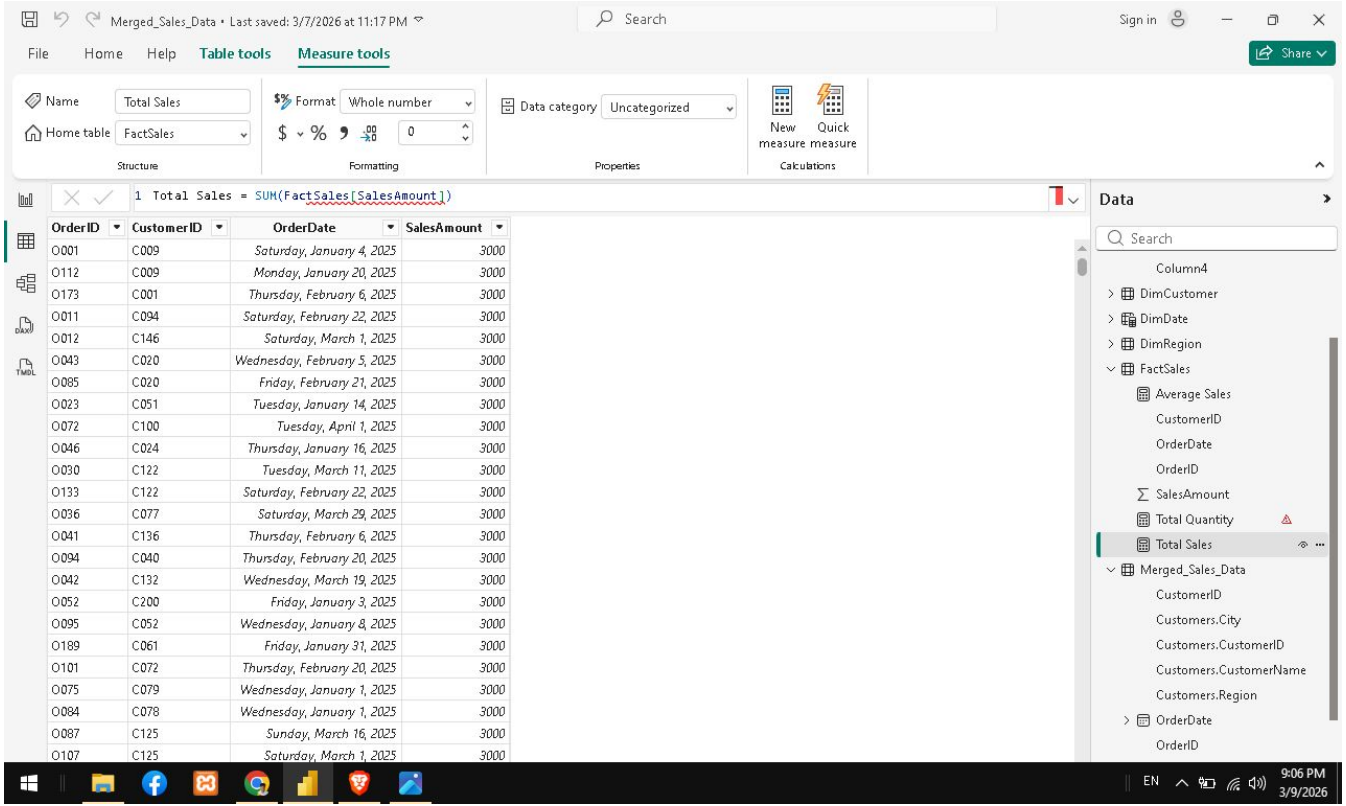


Enhancement Activities | Enhancement 3: Compare Flat vs Star Schema Performance

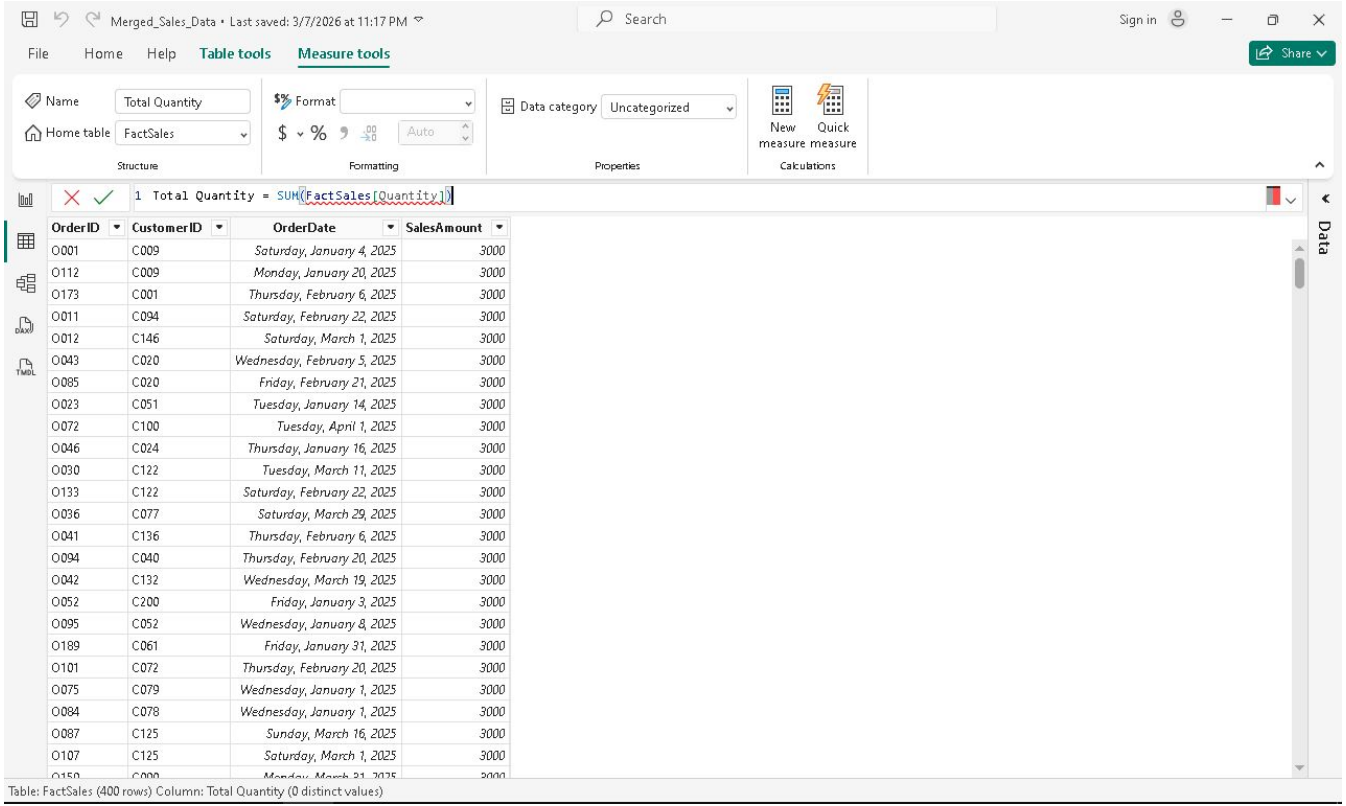
Create visuals using a star schema



Introduction to DAX | Create Measure Step 1: Total Sales



Introduction to DAX | Create Measure Step 2: Total Quantity



OrderID	CustomerID	OrderDate	SalesAmount
O001	C009	Saturday, January 4, 2025	3000
O112	C009	Monday, January 20, 2025	3000
O173	C001	Thursday, February 6, 2025	3000
O011	C094	Saturday, February 22, 2025	3000
O012	C146	Saturday, March 1, 2025	3000
O043	C020	Wednesday, February 5, 2025	3000
O085	C020	Friday, February 21, 2025	3000
O023	C051	Tuesday, January 14, 2025	3000
O072	C100	Tuesday, April 1, 2025	3000
O046	C024	Thursday, January 16, 2025	3000
O030	C122	Tuesday, March 11, 2025	3000
O133	C122	Saturday, February 22, 2025	3000
O036	C077	Saturday, March 29, 2025	3000
O041	C136	Thursday, February 6, 2025	3000
O094	C040	Thursday, February 20, 2025	3000
O042	C132	Wednesday, March 19, 2025	3000
O052	C200	Friday, January 3, 2025	3000
O095	C052	Wednesday, January 8, 2025	3000
O189	C061	Friday, January 31, 2025	3000
O101	C072	Thursday, February 20, 2025	3000
O075	C079	Wednesday, January 1, 2025	3000
O084	C078	Wednesday, January 1, 2025	3000
O087	C125	Sunday, March 16, 2025	3000
O107	C125	Saturday, March 1, 2025	3000

Introduction to DAX | Create Measure Step 2: Average Sales

Merged_Sales_Data • Last saved: 3/7/2026 at 11:17 PM

File | Home | Help | Table tools | Measure tools

Name: Average Sales
Home table: FactSales
Structure

Format: General
Auto
Formatting

Data category: Uncategorized
Properties

New measure | Quick measure
Calculations

```
1 Average Sales = AVERAGE(FactSales[SalesAmount])
```

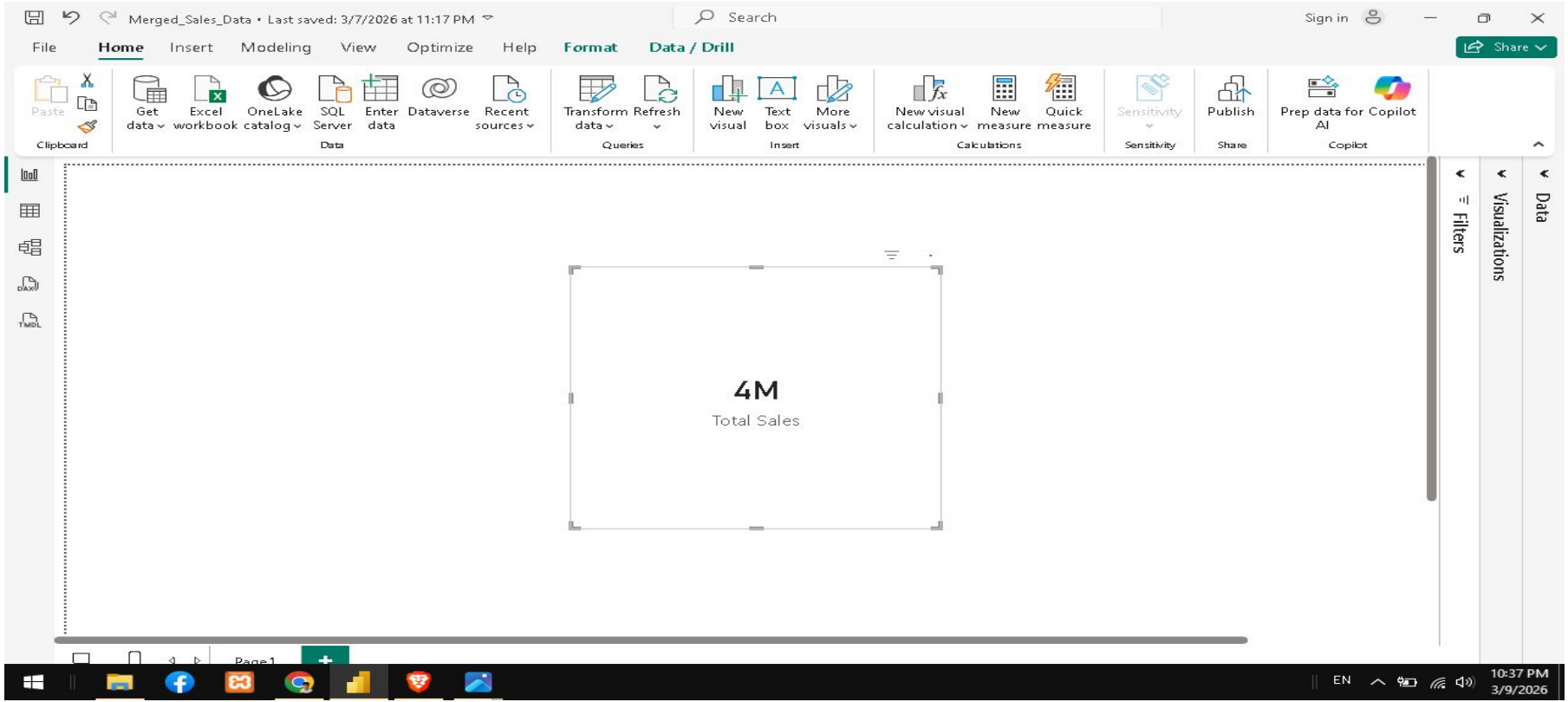
OrderID	CustomerID	OrderDate	SalesAmount
O001	C009	Saturday, January 4, 2025	3000
O112	C009	Monday, January 20, 2025	3000
O173	C001	Thursday, February 6, 2025	3000
O011	C094	Saturday, February 22, 2025	3000
O012	C146	Saturday, March 1, 2025	3000
O043	C020	Wednesday, February 5, 2025	3000
O085	C020	Friday, February 21, 2025	3000
O023	C051	Tuesday, January 14, 2025	3000
O072	C100	Tuesday, April 1, 2025	3000
O046	C024	Thursday, January 16, 2025	3000
O030	C122	Tuesday, March 11, 2025	3000
O133	C122	Saturday, February 22, 2025	3000
O036	C077	Saturday, March 29, 2025	3000
O041	C136	Thursday, February 6, 2025	3000
O094	C040	Thursday, February 20, 2025	3000
O042	C132	Wednesday, March 19, 2025	3000
O052	C200	Friday, January 3, 2025	3000
O095	C052	Wednesday, January 8, 2025	3000
O189	C061	Friday, January 31, 2025	3000
O101	C072	Thursday, February 20, 2025	3000
O075	C079	Wednesday, January 1, 2025	3000
O084	C078	Wednesday, January 1, 2025	3000
O087	C125	Sunday, March 16, 2025	3000
O107	C125	Saturday, March 1, 2025	3000

Data

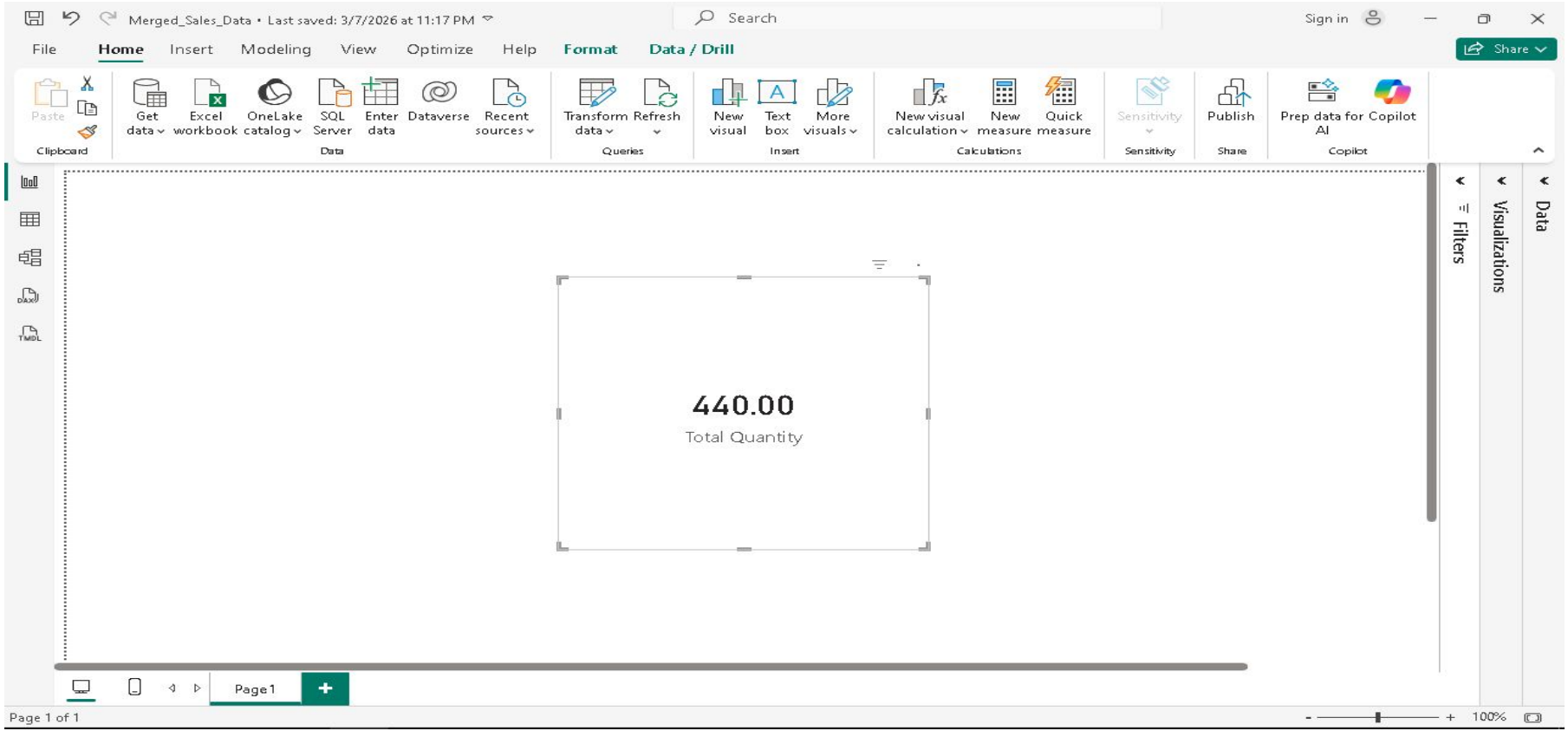
- Column4
- DimCustomer
- DimDate
- DimRegion
- FactSales
  - Average Sales
  - CustomerID
  - OrderDate
  - OrderID
  - SalesAmount
  - Total Quantity
  - Total Sales
- Merged_Sales_Data
  - CustomerID
  - Customers.City
  - Customers.CustomerID
  - Customers.CustomerName
  - Customers.Region
  - OrderDate
  - OrderID

EN 9:04 PM 3/9/2026

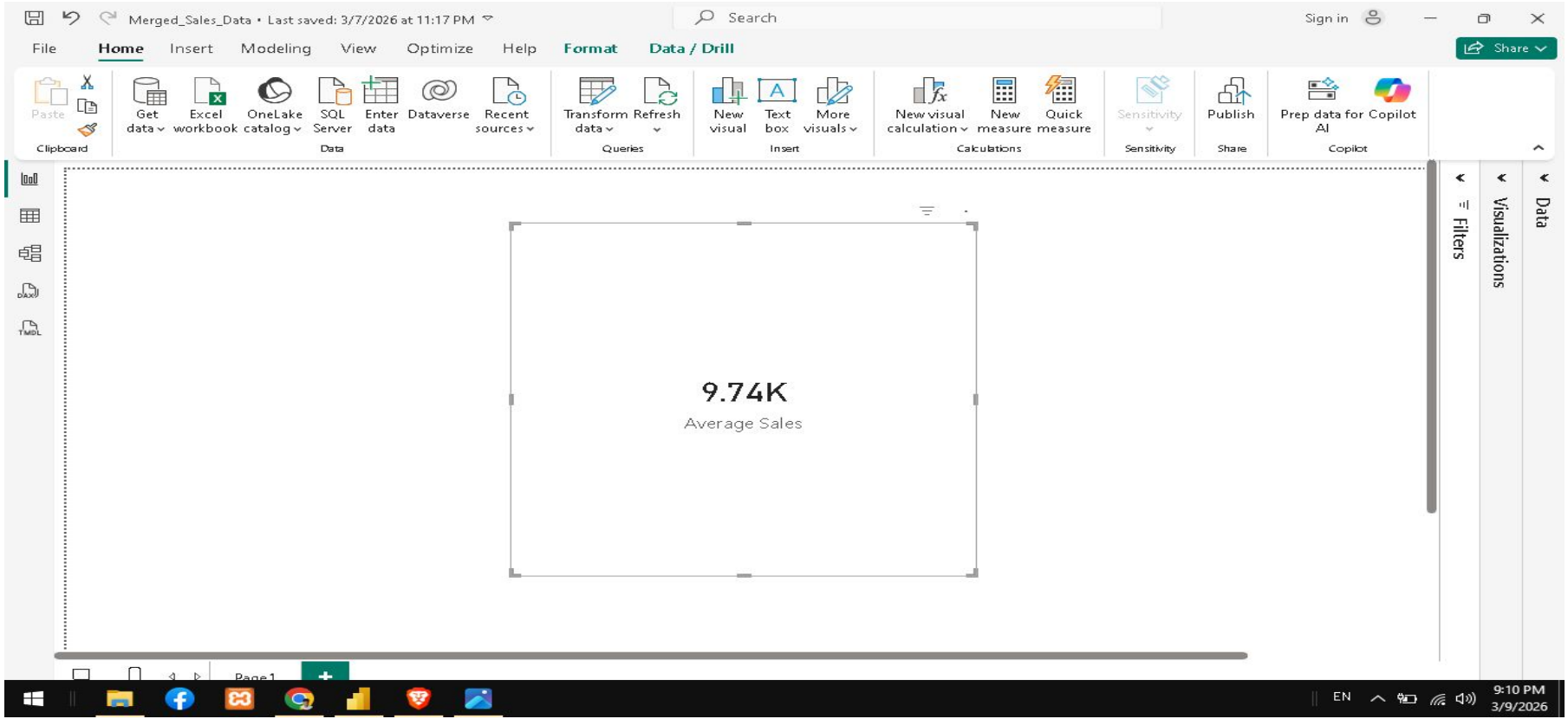
Display Measures



Display Measures



Display Measures



Guide Questions

1. What is a DAX measure?

A DAX measure is a calculation created in Power BI that evaluates dynamically based on the context of your report. Unlike a column that exists in the table, a measure does not store data in each row; instead, it calculates values on the fly depending on filters applied in visuals such as tables, charts, or cards. For example, a measure like Total Sales = SUM(FactSales[SalesAmount]) computes the total sales for whatever subset of data is currently displayed, making measures highly flexible and context-aware.

Guide Questions

2. Difference between SUM and AVERAGE

The SUM function adds all the values in a column to produce a total, whereas the AVERAGE function calculates the mean of the values, dividing the total by the number of rows. For example, SUM(FactSales[SalesAmount]) returns the total sales amount across all rows, while AVERAGE(FactSales[SalesAmount]) gives the typical sale amount per row. While both are aggregation functions, SUM is used to find totals, and AVERAGE is used to understand the general trend or typical value.

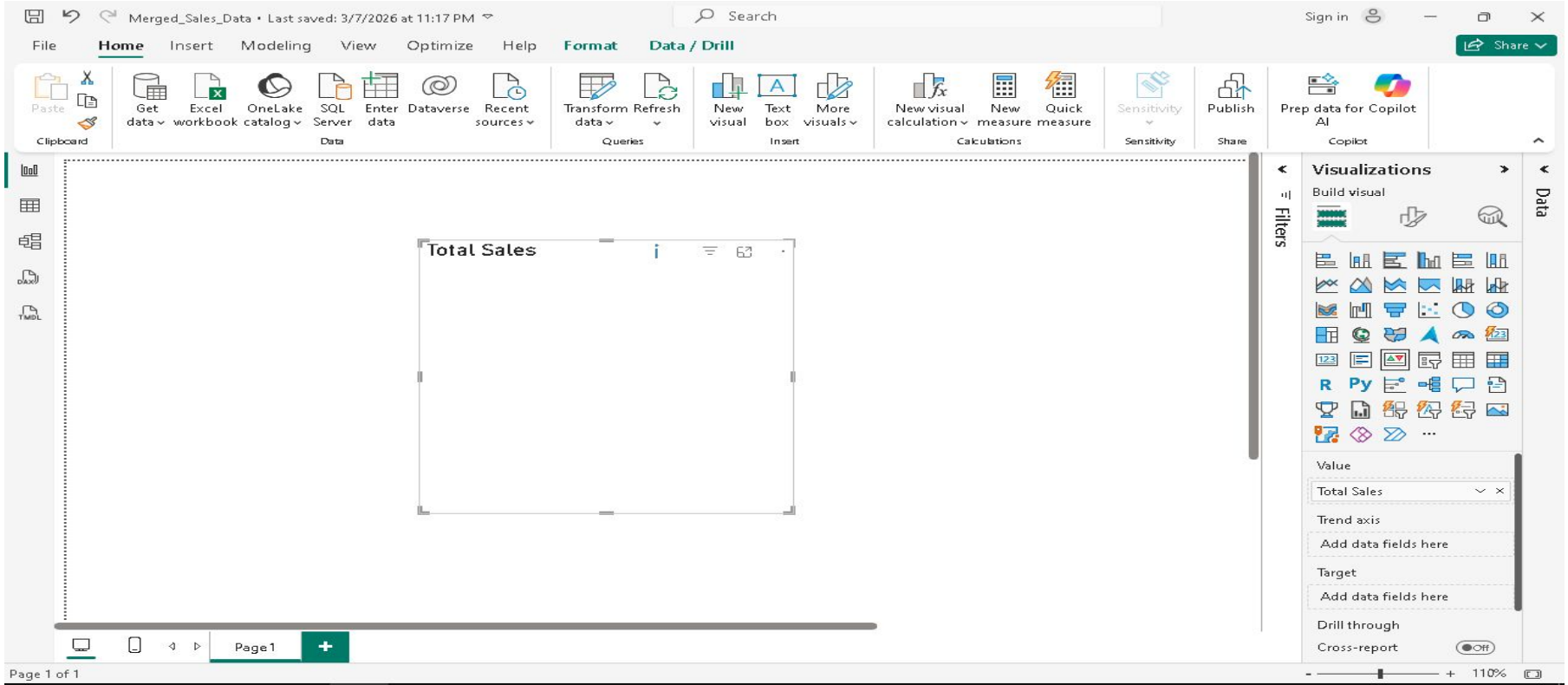
Guide Questions

3. Why use measures instead of calculated columns?

Measures are preferred when performing dynamic aggregations because they calculate results on the fly and do not increase the size of your data model. They respond to filters and slicers in reports, providing context-aware calculations such as totals, averages, or ratios. Calculated columns, on the other hand, store a value for each row in the table and are static after data refresh. While calculated columns are useful for row-level calculations, measures are more efficient for aggregations and interactive reporting, making them ideal for summaries and analysis in Power BI.

Enhancement Activities

Create a KPI visual using Total Sales.



Enhancement Activities

Compare measure vs calculated column performance:

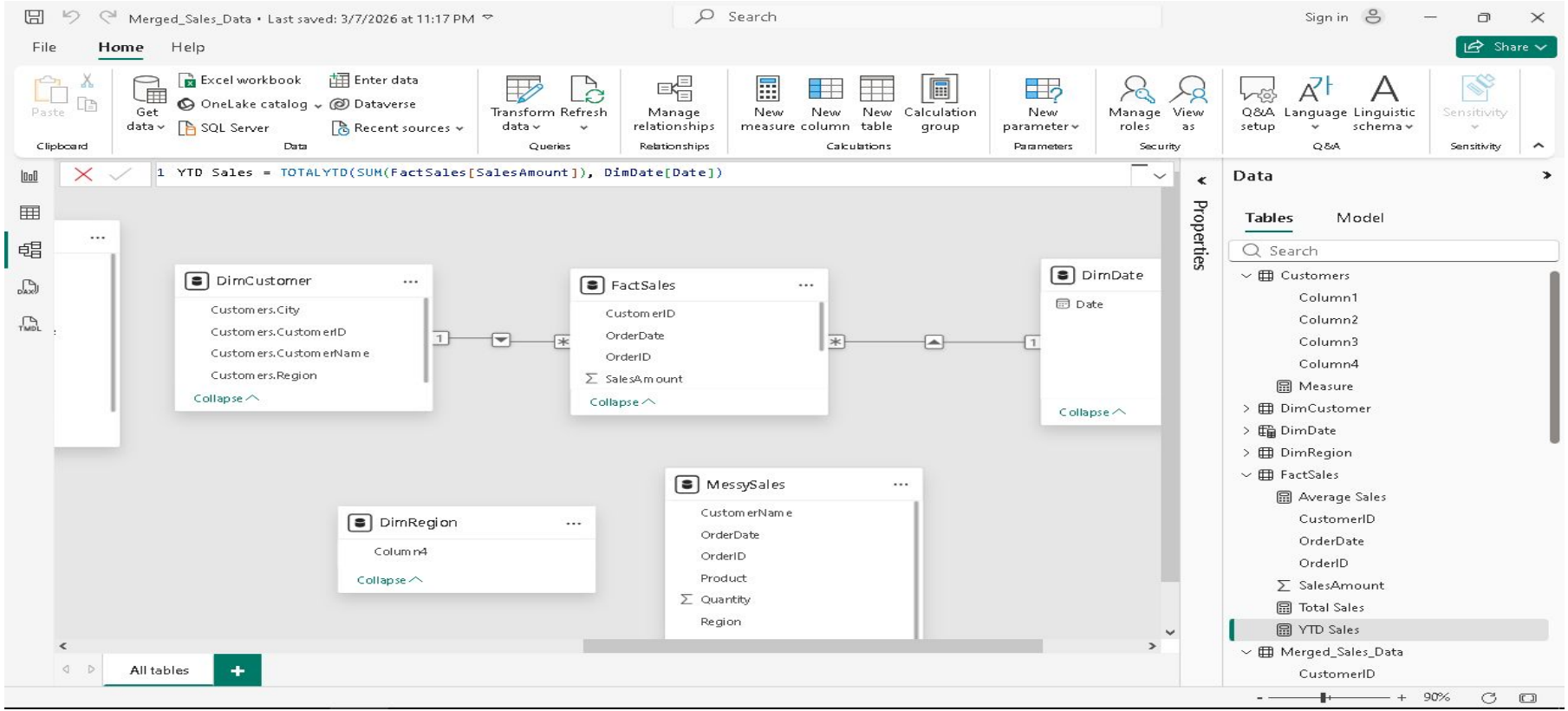
Measures and calculated columns both perform calculations in Power BI, but they behave differently in terms of performance. A calculated column computes a value for every row in the table and stores it in the data model, which can increase memory usage and slow down the model for very large datasets. In contrast, a measure calculates its result on the fly only when it is used in a visual, and it dynamically responds to filters or slicers applied in the report. Because measures do not store data for every row, they are generally more efficient and better for aggregations over large tables, whereas calculated columns are useful when a row-level value is needed for further calculations or relationships.

Enhancement Activities

Explore COUNT vs COUNTROWS:

Both COUNT and COUNTROWS are aggregation functions in DAX, but they serve slightly different purposes. COUNT counts the number of non-blank values in a single column, so if a column contains blank entries, they are ignored in the count. COUNTROWS, on the other hand, counts the total number of rows in a table or a filtered table, regardless of the values in any specific column. For example, COUNT(FactSales[Quantity]) counts how many rows have a value in the Quantity column, while COUNTROWS(FactSales) counts all rows in the FactSales table, including rows where Quantity may be blank. Choosing between them depends on whether you need to count specific column values or all rows in the table.

DAX Functions | Step 1: Year-to-Date Sales



DAX Functions | Step 2: Monthly Sales Growth

Merged_Sales_Data • Last saved: Today at 11:30 PM

```
Monthly Growth =
DIVIDE(
[Total Sales] - CALCULATE([Total Sales], PREVIOUSMONTH(DimDate[Date])),
CALCULATE([Total Sales], PREVIOUSMONTH(DimDate[Date])))
```

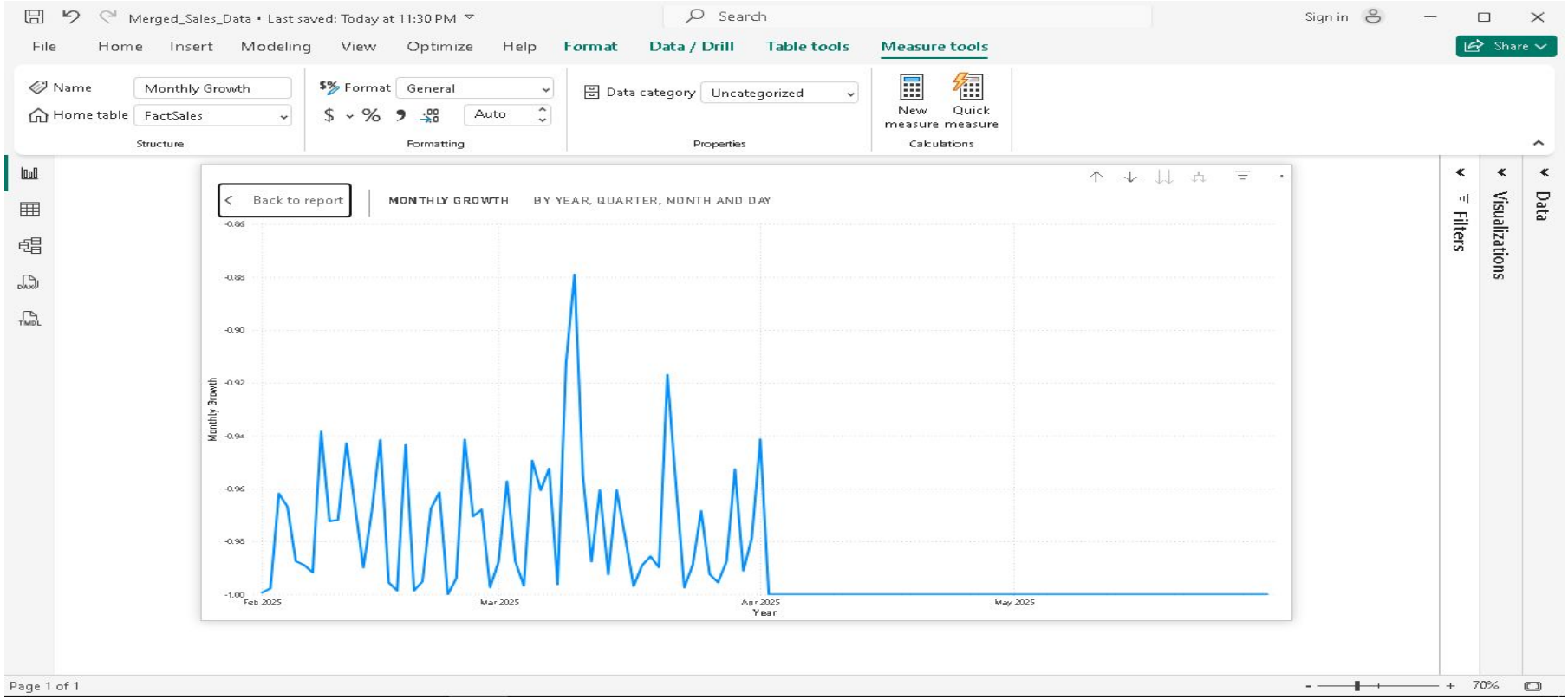
Customers
Column1
Column2
Column3
Column4
Collapse

Orders
CustomerID
Customers.Column1
Customers.Column2
Customers.Column3
Collapse

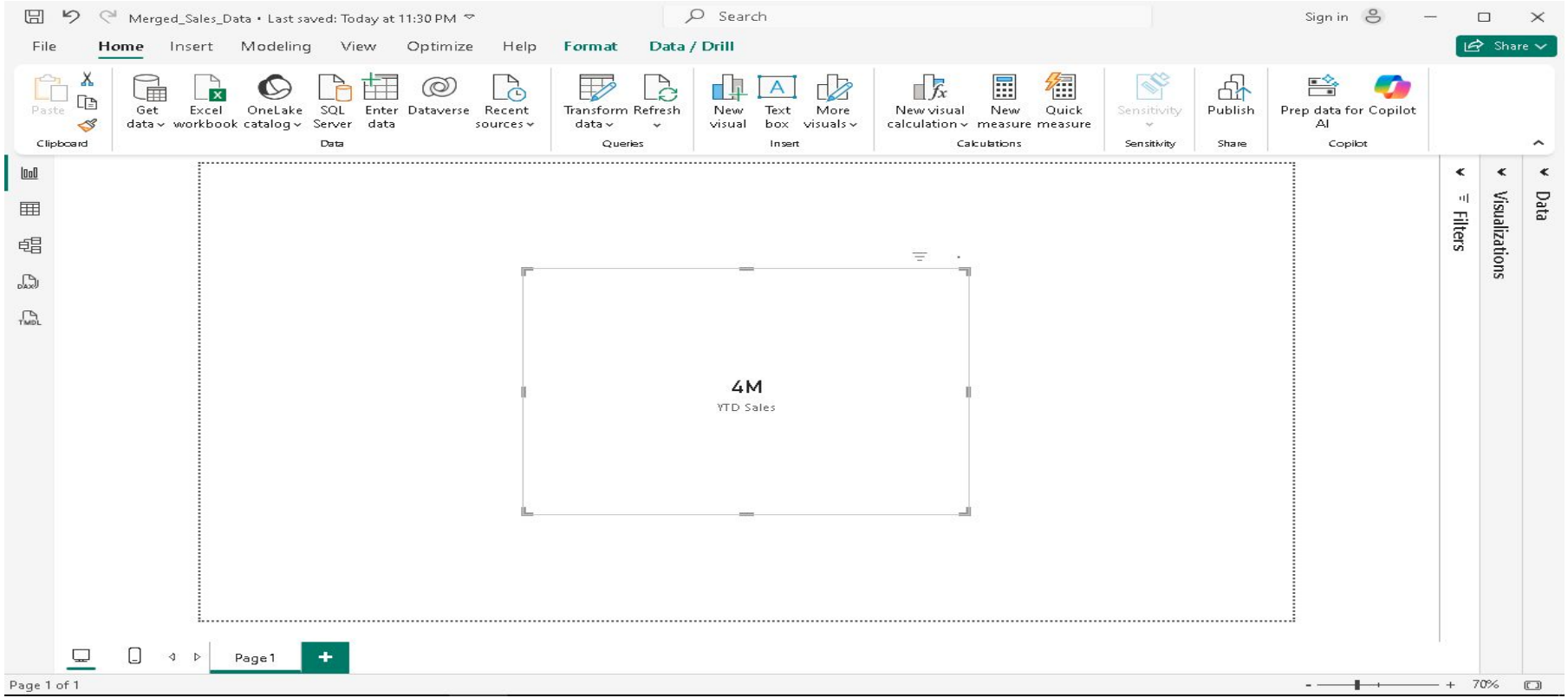
Customers.City
Customers.CustomerID
Customers.CustomerName
Customers.Region
OrderDate
OrderID
SalesAmount
Collapse

Data
Tables
Model
Customers
Column1
Column2
Column3
Column4
Monthly Growth
DimCustomer
DimDate
DimRegion
FactSales
Merged_Sales_Data
MessySales
Orders

DAX Functions | Create Visualization: Line chart showing monthly sales



DAX Functions | Create Visualization: Card showing YTD Sales



Guide Questions

1. What is time intelligence in DAX?

Time intelligence in DAX refers to a set of functions that allow you to perform calculations based on dates and periods. These functions make it easy to analyze trends over time, such as comparing sales month over month, calculating year-to-date totals, or determining the difference between periods. Time intelligence simplifies complex date calculations, enabling insights like growth rates, cumulative totals, or rolling averages without manually writing complex formulas for each period.

Guide Questions

2. Why is a date table required?

A date table is required in Power BI to enable accurate time-based calculations. It provides a continuous list of dates along with attributes such as year, month, quarter, and day, which DAX functions use to aggregate or filter data over time. Without a proper date table, time intelligence functions may produce incorrect results, especially when dealing with missing dates, gaps in data, or fiscal calendars. The date table acts as a reference for DAX to understand the chronological relationships between data points.

Guide Questions

3. How does PREVIOUSMONTH help analyze trends?

The PREVIOUSMONTH function in DAX allows you to compare a value for the current month against the previous month. For example, it can calculate last month’s sales, which is useful for identifying trends, growth, or decline in performance. By providing a direct reference to the prior period, PREVIOUSMONTH simplifies trend analysis, helping decision-makers quickly see whether metrics like revenue, quantity sold, or expenses are increasing or decreasing compared to the preceding month.

Guide Questions

4. What insights can YTD Sales provide?

Year-to-date (YTD) Sales measures cumulative sales from the beginning of the year up to the current date. YTD analysis provides insights into overall performance relative to annual goals, seasonal patterns, or progress over time. It helps businesses understand whether they are on track to meet targets, identify periods of strong or weak performance, and make informed decisions about resource allocation, marketing campaigns, or sales strategies throughout the year.

Enhancement Activities | Create Quarter-to-Date Sales.

Merged_Sales_Data • Last saved: Yesterday at 11:30 PM

File | Home | Help | Table tools | Measure tools

Name: QTD Sales
Home table: FactSales
Structure

Format: Whole number
Formatting

Data category: Uncategorized
Properties

New measure | Quick measure
Calculations

```
QTD Sales =
TOTALQTD(
    SUM(FactSales[SalesAmount]),
    DimDate[Date]
)
```

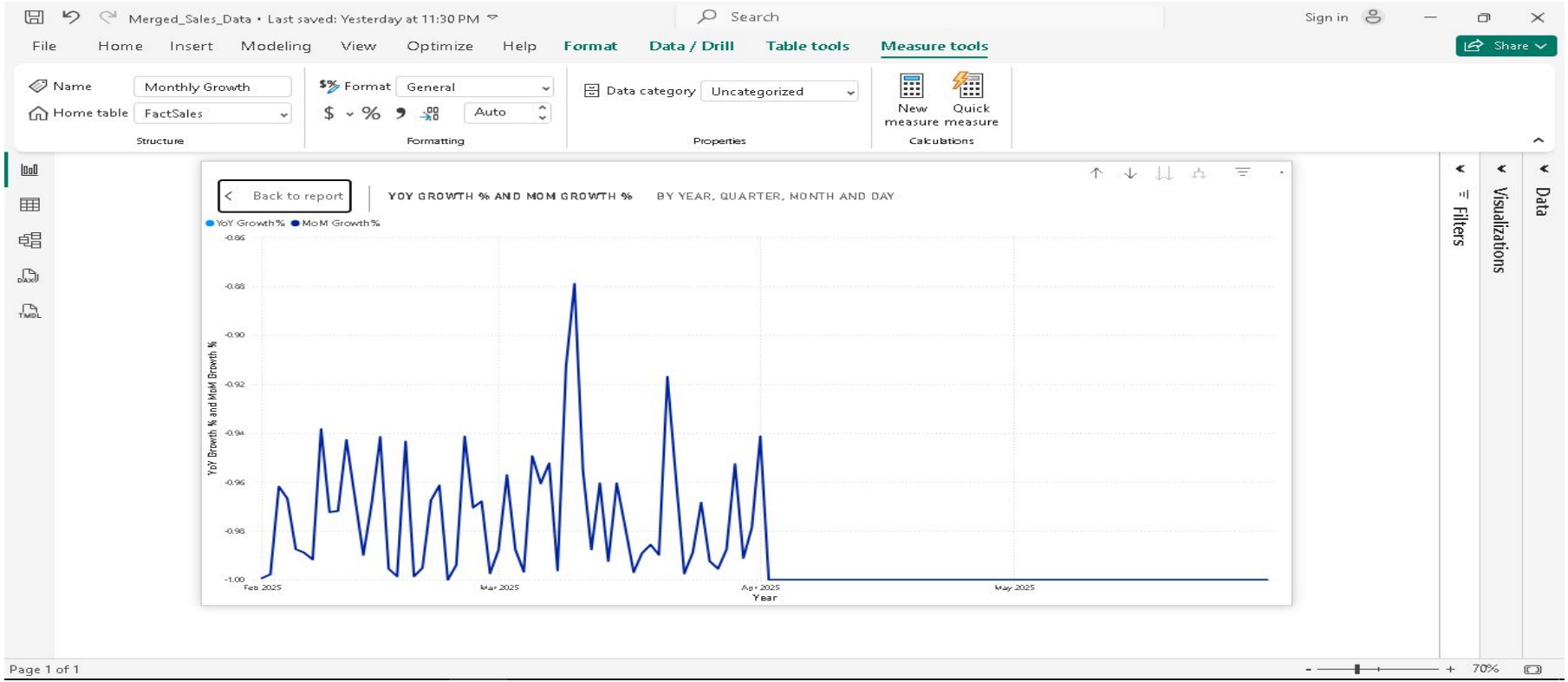
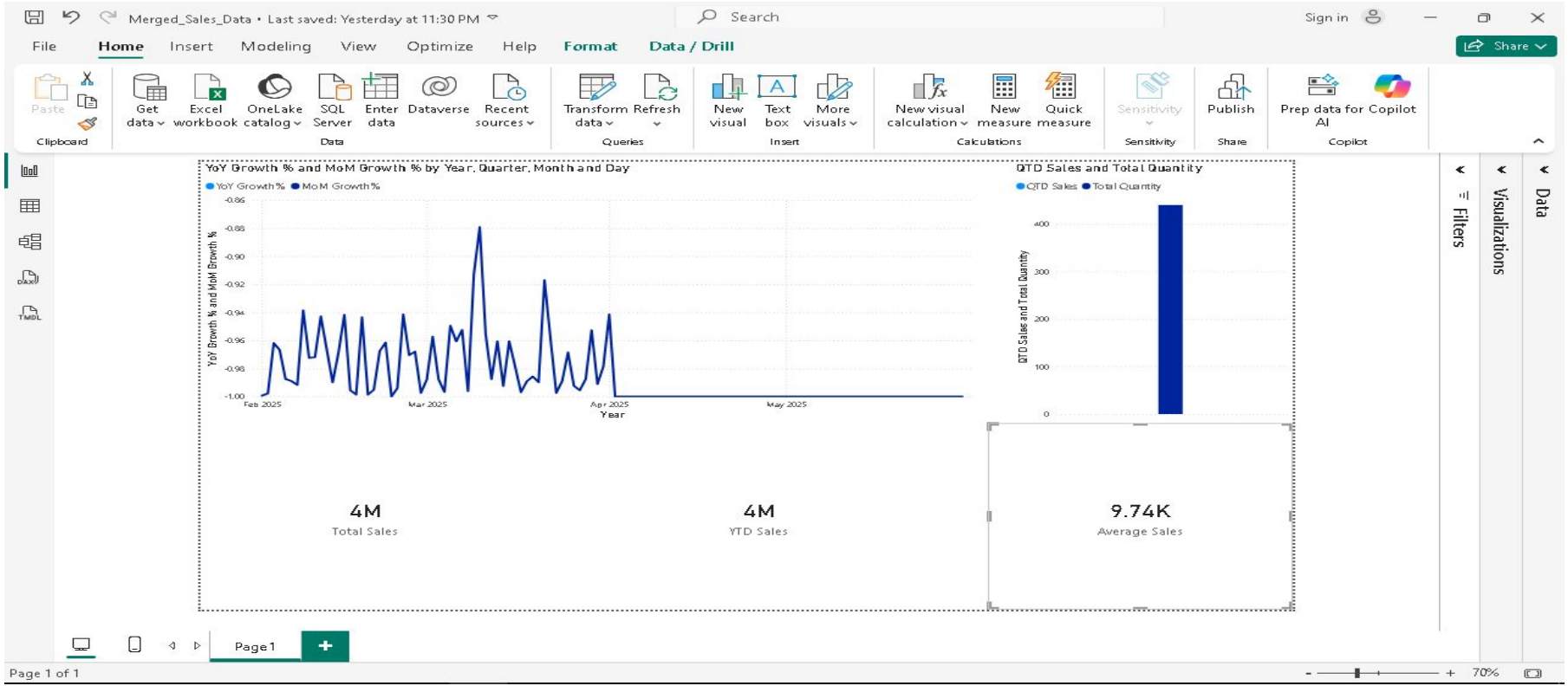
OrderID	CustomerID	OrderDate	SalesAmount
O001	C009	Saturday, January 4, 2025	3000
O112	C009	Monday, January 20, 2025	3000
O173	C001	Thursday, February 6, 2025	3000
O011	C094	Saturday, February 22, 2025	3000
O012	C146	Saturday, March 1, 2025	3000
O043	C020	Wednesday, February 5, 2025	3000
O085	C020	Friday, February 21, 2025	3000
O023	C051	Tuesday, January 14, 2025	3000
O072	C100	Tuesday, April 1, 2025	3000
O046	C024	Thursday, January 16, 2025	3000
O030	C122	Tuesday, March 11, 2025	3000
O133	C122	Saturday, February 22, 2025	3000
O036	C077	Saturday, March 29, 2025	3000
O041	C136	Thursday, February 6, 2025	3000
O094	C040	Thursday, February 20, 2025	3000
O042	C132	Wednesday, March 19, 2025	3000
O052	C200	Friday, January 3, 2025	3000
O095	C052	Wednesday, January 8, 2025	3000
O189	C061	Friday, January 31, 2025	3000
O101	C072	Thursday, February 20, 2025	3000
O075	C079	Wednesday, January 1, 2025	3000

Data

- Customers
  - Column1
  - Column2
  - Column3
  - Column4
- DimCustomer
- DimDate
- DimRegion
- FactSales
  - Average Sales
  - CustomerID
  - MoM Growth %
  - Monthly Growth
  - OrderDate
  - OrderID
  - QTD Sales
  - SalesAmount
  - Total Sales
  - YoY Growth %
  - YTD Sales

Table: FactSales (400 rows) Column: QTD Sales (0 distinct values)

Enhancement Activities | Compare Month-over-Month vs Year-over-Year growth.



Enhancement Activities | Build a dashboard using all measures.